Automatic Umbrella System

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Overview and Reflection

Introduction

Purpose

- Automatic umbrella
- Rain intensity indicator

Humanity Efforts

- Helps individuals who can't open their umbrella on their own.
- Protects individuals from being soaked in rain.



Inputs and outputs

Inputs:

Humidity sensor

Contact switch

Output:
Motor
Flashing LED



Constraints and Requirements

Constraints:

Limited to the materials offered such as cardboard and duct tape

Needed to create an umbrella light enough for the motor to move using the materials we had.

Requirements:
Needed humidity sensor, LED, and at least one other switch.

Use of AI code

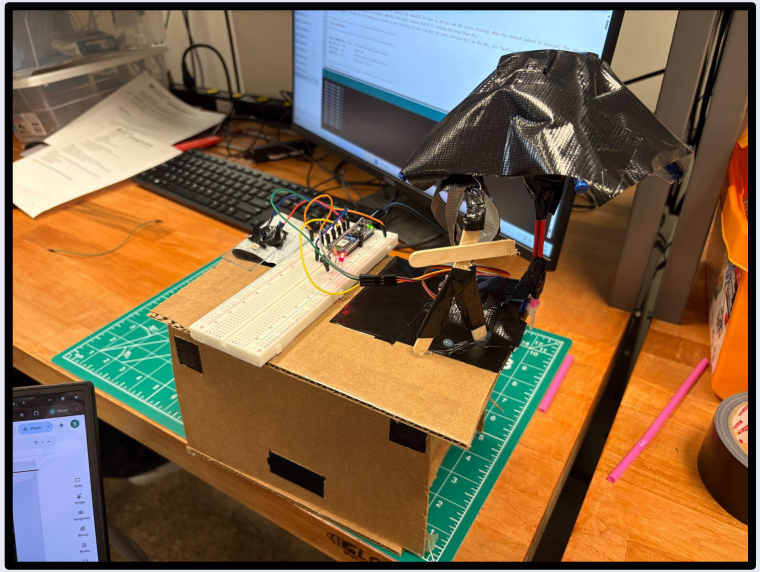
Implementation of Design Process

Research

- Examined the basic design of an Umbrella so it could be recreated with present materials

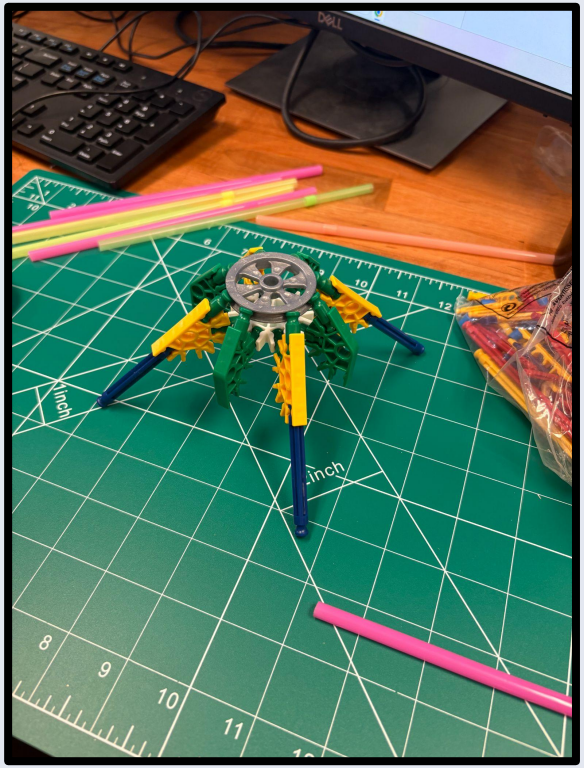
Prototype

- Utilized knex, duct tape, and servo motor for main umbrella component with our sensor connected to detect moisture.

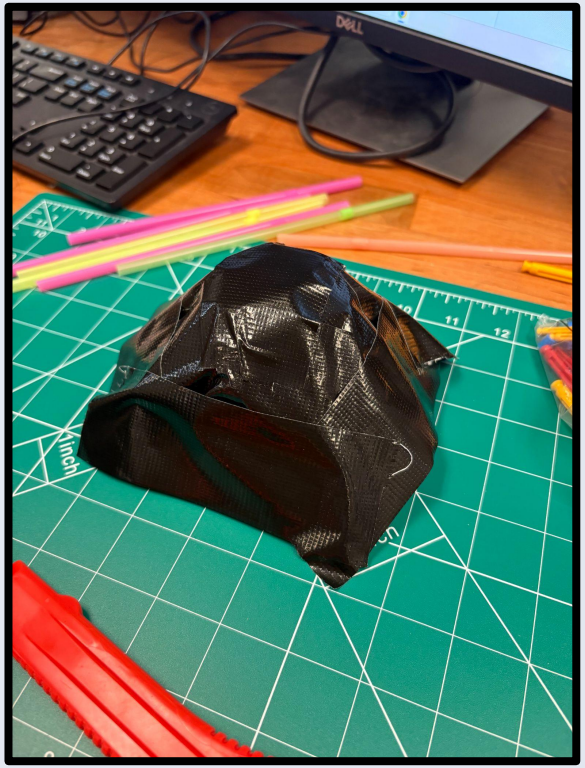


Pictured Above is Prototype

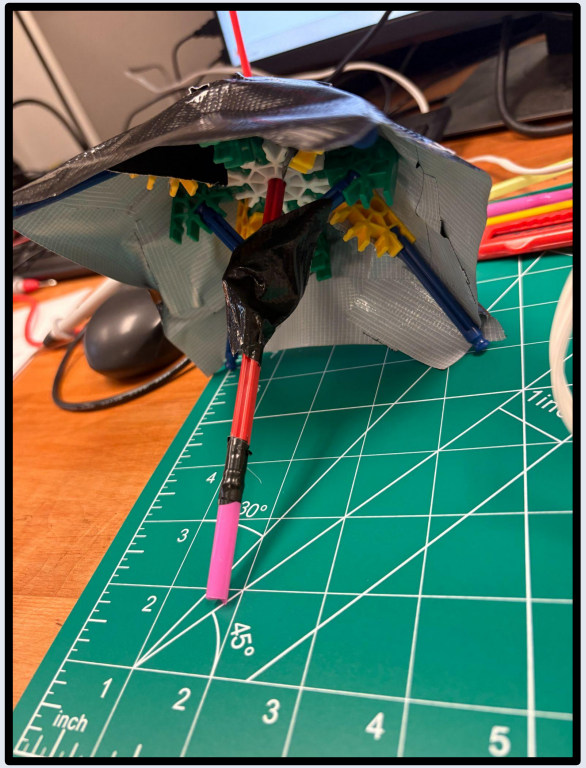
Stages of Prototype



Pictured above is early stage prototype



Pictured above is middle stage prototype



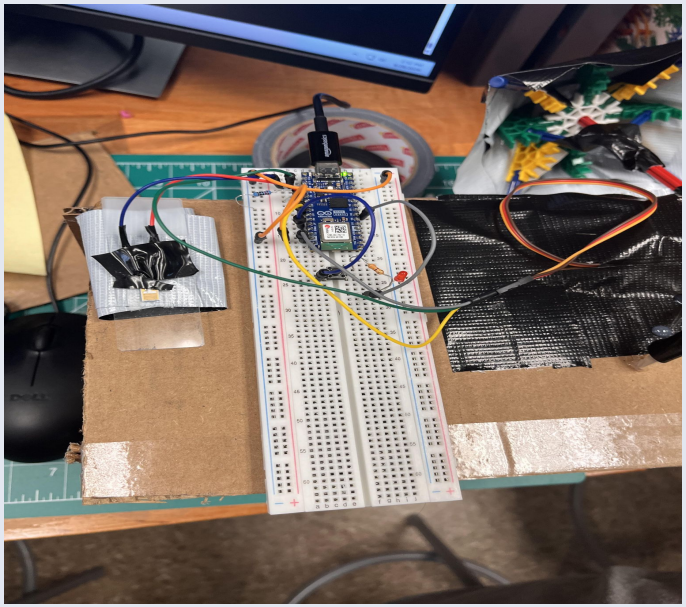
Pictured above is final stages prototype

Implementation of Wiring & Coding

Pins

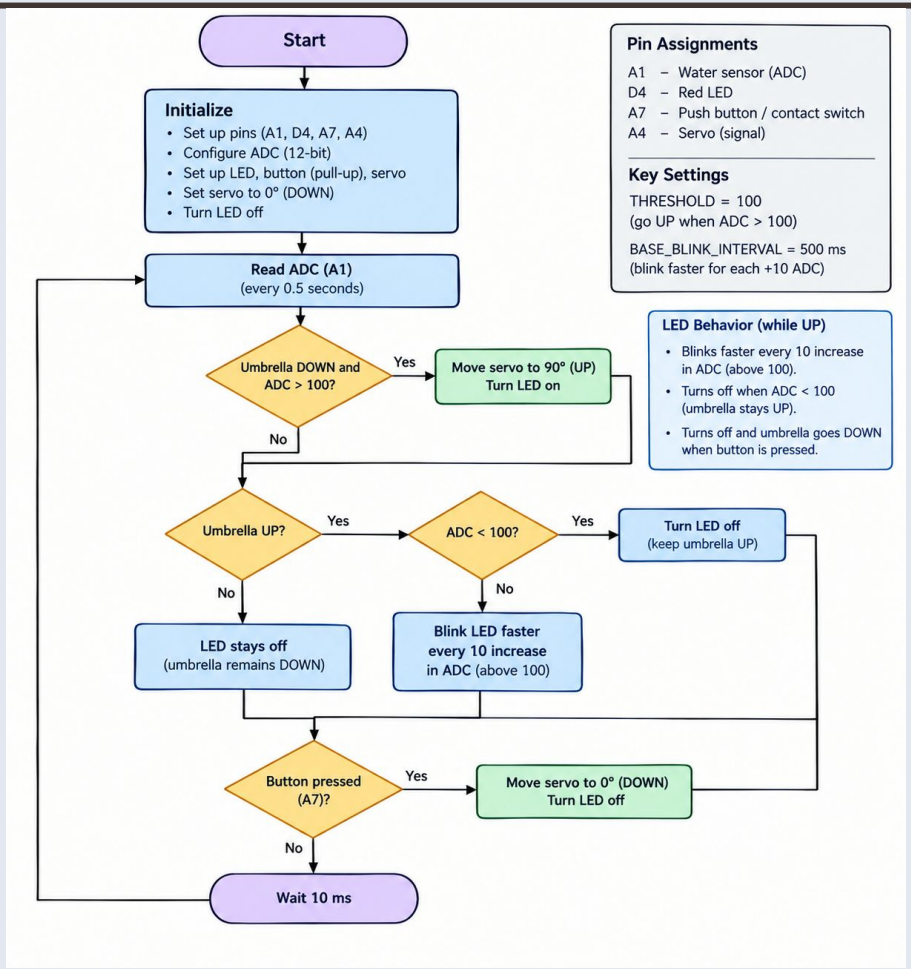
- Humidity sensor (A1)
- Red LED (D4)
- Contact Switch (A7)
- Servo Motor (A4)

Connections



Pictured Above is Prototype

Flow


Start
Initialize
• Set up pins (A1, D4, A7, A4)
• Configure ADC (12-bit)
• Set up LED, button (pull-up), servo
• Set servo to 0° (DOWN)
• Turn LED off
Read ADC (A1)
(every 0.5 seconds)
Umbrella DOWN and ADC > 100?
Yes
Move servo to 90° (UP)
Turn LED on
No
Umbrella UP?
Yes
ADC < 100?
Yes
Turn LED off
(keep umbrella UP)
No
LED stays off
(umbrella remains DOWN)
No
Blink LED faster every 10 increase in ADC (above 100)
Button pressed (A7)?
Yes
Move servo to 0° (DOWN)
Turn LED off
No
Wait 10 ms
Pin Assignments
A1 – Water sensor (ADC)
D4 – Red LED
A7 – Push button / contact switch
A4 – Servo (signal)
Key Settings
THRESHOLD = 100
(go UP when ADC > 100)
BASE_BLINK_INTERVAL = 500 ms
(blink faster for each +10 ADC)
LED Behavior (while UP)
• Blinks faster every 10 increase in ADC (above 100).
• Turns off when ADC < 100 (umbrella stays UP).
• Turns off and umbrella goes DOWN when button is pressed.

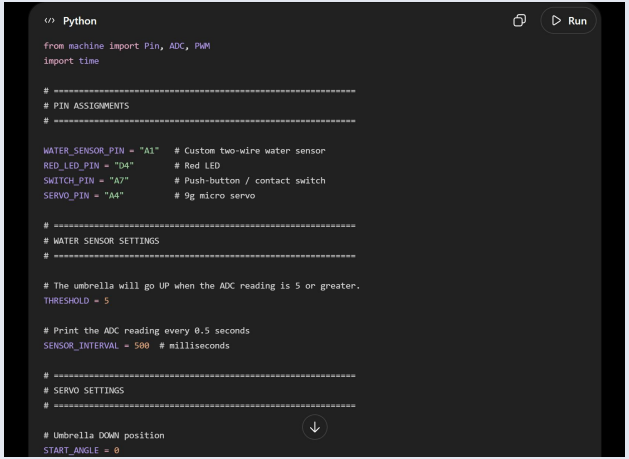
Chart

AI Usage

Creating the Code

Breadboard Setup Assistance

Revising the Code



Results

Sensor Input

Motor Output

LED Output

Manual Off Switch



Conclusion

Accomplishments

- Successfully created a functioning umbrella that opens automatically using one sensor and closes using another sensor.

Functionality

- Umbrella shielded the test subject.

Demonstration Video:

